HYDROWORX

500

Owner's Manual



HYDROWORX
1420 Stoneridge Drive, Suite C, Middletown PA 17057
717-902-1923
HydroWorx.com

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1.0 Disclaimer and Confidentiality Notice

This confidential Owner's Manual (the "Manual") is delivered to parties (the "Recipient") who have purchased a HydroWorx System from HydroWorx International, Inc. This Manual is highly confidential and proprietary. Receipt of the Manual is solely in connection with the Recipient's consideration of the enclosed information. Use of this Manual for any other purpose is not authorized.

By accepting this Manual, the Recipient agrees to keep confidential the information contained herein or made available by the Company in connection with any service, maintenance, or repair of the HydroWorx System. By accepting this Manual, the Recipient also agrees to make no copies of the Manual in whole, in part, or in any form and to use this Manual solely for the purposes of understanding the proper use of the HydroWorx System.

2.0 Product Description

The HydroWorx 500 is a permanently connected hydrotherapy system. It incorporates a one-piece tub, integrated treadmill, touchscreen controls, therapy jets, and a water recirculation system (pump, filter, heater, sanitizer).

The tub is constructed of acrylic with fiberglass and wooden reinforcement.

The treadmill is constructed of stainless steel and is integrated into the floor of the structure which results in a flush finish. The treadmill is electro-mechanically driven (no hydraulics). The treadmill can accommodate a patient and a therapist with a maximum combined body weight of 550lb. The treadmill speed is adjustable from 0.2 – 7.5 mph.

The mechanical and electrical pool components are typically located in the lower level mechanical room or in an adjacent equipment room.

The 500 can be controlled via an HMI (Human Machine Interface) and a tablet. The HMI is mounted in the pool room in front of the 500. A tablet is provided for remote control.

Specifications	
Weight (empty)	1800 lb
Weight (full) without bathers	18,000 lb
Water surface dimensions	7' x 13'-6"
Water depth, deep end	4'-6"
Water depth, shallow end	3'-8"

Water volume	2,000 gal
Recirculation flow rate	20-60 gpm
Front jets flow rate	400 gpm
Side jet flow rate	110 gpm
Variable water depth change	10”

3.0 Limited Warranty

HydroWorx International, Inc. (HydroWorx) warrants solely to the original Buyer that this product, consisting of the pool shell, treadmill, jet system, electronic controls, and recirculation equipment will be free from defects in materials and workmanship, when given normal, proper and intended installation and usage, for the period set forth below from the date noted on signed start-up completion form.

LIMITED WARRANTY PERIOD	COMPONENTS COVERED
One (1) Year	▪ Water Circulation Equipment* (heater, filter housing, erosion feeder and pump) ▪ Therapy Jet and Pumps ▪ Treadmill System* (Motor and drive components, bearings, seals, belt) ▪ Control System ▪ HMI System ▪ Water transfer system (pump, valves, storage tank) ▪ Video System ▪ Ceramic tile
Five (5) Year	▪ Pool Shell ▪ Treadmill frame structural integrity

*Proper maintenance must be followed to keep the warranty intact.

During the limited warranty period, HydroWorx will repair or replace, at its option and without cost to the Buyer, any portion of the product which is confirmed by HydroWorx to be defective.

This warranty shall not apply to any expendable component, such as, but not limited to chemicals, fuses, filters, and lubricants. HydroWorx shall have no obligations to make repairs or replacements which result, in whole or in part, from (I) catastrophe, fault or negligence of Buyer, (II) improper or unauthorized use of product, (III) maintenance or installation of the product which is not in full compliance with the requirements of the HydroWorx Owner's Manual and Design and Installation Guide, (IV) the presence of debris or silt in the product, (V) the failure to maintain water cleanliness and chemistry, accordance with the recommendations of recirculation equipment suppliers, (VI) any modification of the product (including, but not limited to, removal or affixing any object to the equipment or removing any component part of the product without prior specific written instruction from HydroWorx), or (VII) causes external to the product such as, but not limited to, power failure.

In the event after inspection by HydroWorx the product proves not to be defective or is not subject to the warranty protection because of normal use, wear and tear, misuse, lapse of time, accident, modification, or any other exclusion specified above, the Buyer will be charged for repair or replacement services at HydroWorx's standard rates then in effect together with the "portal to portal" expenses incurred by HydroWorx in providing the service.

The provisions of the foregoing warranty are in lieu of any other warranty, whether express or implied, written or oral, including any warranty of merchantability or fitness for a particular purpose. HydroWorx's liability arising out of the manufacture, sale or supplying of the products or its use or disposition whether based upon warranty, contract, tort or otherwise, shall not exceed the actual purchase price paid by the buyer for the product. In no event shall HydroWorx be liable to buyer or any other person or entity including, but not limited to, loss of profits or loss of use of damages arising out of the manufacture, sale or supplying of the product the foregoing warranty applies to the buyer only and shall not extend to any other person or entity, including, without limitation, customers of the buyer.

Some states do not allow limitations on how long an implied warranty lasts, or the exclusion or limitation of incidental or consequential damages, so the above limitation or exclusion may not apply to some buyers. This warranty gives the buyer specific legal rights, and the buyer may also have other rights which vary from state to state.

Warning: users should not use this equipment without constant adult supervision from a qualified practitioner who has been trained on the use of the HydroWorx system. Users should consult his/her physician before using this equipment. At all times HydroWorx disclaims and shall not be liable for any personal injury sustained by any user whether authorized or unauthorized, of the product.

Any modification to the as-delivered and installed unit that is not made by the factory or factory authorized personnel shall be interpreted as an unauthorized modification. Any unauthorized modification that is not reviewed and approved by HydroWorx prior to making such modification of the unit or any other device sold by or through HydroWorx shall release HydroWorx of any and all liability for personal injury or property damage relating to or arising from use with the modified product. Any modification to a HydroWorx product not approved by HydroWorx shall void any and all warranty of the product.

The unit must be installed in accordance with the 500 Design & Installation Guide. Any variance not approved by HydroWorx may result in a voided warranty.

4.0 Safety

IMPORTANT SAFETY INSTRUCTIONS.

READ AND FOLLOW ALL INSTRUCTIONS.

SAVE THESE INSTRUCTIONS.

READ THIS MANUAL ENTIRELY BEFORE OPERATING THE UNIT.

⚠ DANGER

- **Risk of Accidental Drowning**. Extreme caution must be exercised to prevent unauthorized access by children. To avoid accidents, ensure that children cannot use this device unless they are supervised at all times.

- **Risk of Injury.** The suction fittings in this unit are sized to match the specific water flow created by the pump. Should the need arise to replace the suction fittings or the pump, be sure that the flow rates are compatible.
  Never operate the unit if the suction fittings are broken or missing. Never replace a suction fitting with one rated less than the flow rate marked on the original suction fitting.
- **Risk of Electric Shock.** Install at least 5 feet from all metal surfaces. As an alternative, the unit may be installed within 5 feet of metal surfaces if each metal surface is permanently connected by a minimum 8AWG solid copper conductor to the wire connector on the terminal box that is provided for this purpose.
- **Risk of Electric Shock.** Do not permit any electrical appliance, such as a light, telephone, radio, or television, within 5 feet of the unit.

⚠ WARNING

- For indoor use only. This unit is not intended for outdoor use.
- To reduce the risk of injury
  - The water in the unit should never exceed 104°F. Water temperatures between 100°F and 104°F are considered safe for a healthy adult. Lower water temperatures are recommended for young children and when use exceeds 10 minutes.
  - Since excessive water temperatures have a high potential for causing fetal damage during the early months of pregnancy, pregnant or possibly pregnant women should limit the water temperature to 100°F.
  - Before entering the unit, the user should measure the water temperature since the tolerance of water temperature-regulating devices varies.
  - The use of alcohol, drugs, or medication before or during use of the unit may lead to unconsciousness with the possibility of drowning.
  - Obese persons and persons with a history of heart disease, low or high blood pressure, circulatory system problems, or diabetes should consult a physician before using the unit.
  - Persons using medication should consult a physician before using the unit since some medication may induce drowsiness while other medication may affect heart rate, blood pressure, and circulation.
- **REDUCE THE RISK OF ELECTROCUTION**

- Be sure each unit is protected by a ground fault interrupter breaker and test prior to each use.

- **REDUCE THE RISK OF CHILD DROWNING**
  - To reduce the risk of injury, do not permit children to use this product unless they are closely supervised at all times.

- User's shall always have supervision from a qualified aquatic therapy practitioner or exercise specialist (therapist).
- If the user begins to feel unwell while using the unit it is important to exit the unit immediately.
- If shoes are worn on the treadmill, they must not have shoelaces that may make contact with the treadmill.
- No Diving.

⚠ CAUTION

- Unauthorized service, repair, or replacement of parts may void your warranty and may damage the unit or injure the user or therapist.
- Maintain proper water chemistry. Improper water chemistry may cause skin and eye irritation as well as damage to the unit.
- Do not use the speed controls to stop the Treadmill or the Jets. Make sure you press OFF, STOP or PAUSE. If the Treadmill or Jet are ON with Zero speed for some time, the drives may Fault. STOP SESSION should be pressed at the end of each session to clear all functions.

SAFETY FEATURES

- The system is equipped with an emergency-stop button located on the HMI. In the event of an emergency, pressing this button will disable power to the unit.
- The unit is equipped with GFCI protection.
- Installation shall comply with this document, the 500 Design and Installation Guide, the National Electric Code, and any other local codes.
- A wire connector is provided on this unit to connect a minimum 8AWG solid copper conductor between this unit and any metal equipment, metal enclosures or electrical equipment, metal water pipe, or conduit within 5 feet of the unit.

5.0 Operating Instructions

Install the 500 in accordance with the Design and Installation Guide.

Refer to Appendix A for diagrams showing the various components of the 500.

5.1 Treadmill

The 500 is equipped with a variable speed treadmill. By adjusting the speed, the user/therapist can vary the intensity of the session. The user should stay centered left to right on the treadmill and not put any sideways force on the belt that could cause a belt tracking problem.

To operate the treadmill:

1. It is recommended that the user straddle the treadmill by placing one foot on each side of the treadmill, onto the side guards before turning on the treadmill.
2. To activate the treadmill, press the speed on/off button on the HMI screen. The treadmill will not start moving until a speed is selected.
3. On the HMI screen either select a preset speed or press the speed increase button. The treadmill has a ramp up function so it may take a couple seconds to reach the desired speed. The user can change the range of preset speeds by holding the mode button on the HMI screen to toggle between sport and therapy mode.
4. ⚠ CAUTION It is recommended to let one foot touch the treadmill belt to get a feeling for how fast the user must walk/run before stepping onto the treadmill with both feet.
5. During the session speed may be increased or decreased by either selecting a preset speed or by pressing the speed increase and decrease buttons on the HMI screen.
6. The speed increase and decrease buttons on the HMI screen adjust the speed in 0.1mph increments each time they are pressed. The speed can be increased/decreased outside of the preset range by using the speed increase and decrease buttons.
7. To stop the treadmill there are several ways to do so depending on the reason why the treadmill is to be stopped.
    a. Speed on/off button on the HMI screen: this is the preferred method if the user just wants to stop the treadmill but continue the session.
    b. Pause Session button on the HMI screen: this is the preferred method if the user wants to stop the treadmill and jet but continue the session.

c. Stop Session button on the HMI screen: use this button when the session is over. This button will stop the treadmill and jet and clear the session timer and distance tracker.

8. The treadmill is programmed to automatically shut off after 1 hour of continuous use. If an extended session is desired, the treadmill can be restarted immediately.

The treadmill includes removable handrails. The handrails can be place along the sides of the treadmill or in front of the treadmill. Refer to Appendix A for an illustration of the two handrail configurations.

5.2 Treadmill Dual Front Jet

The 500 is equipped with a dual jet in front of the treadmill that can be used as resistance for running and upper body exercises, or it can be used with the massage hose for muscle massage.

To operate the jet:

1. To activate the jet, press the Jet on/off button on the HMI screen. If starting a new session, the jet will not start until a speed is selected. If the jet has already been used in the current session the jet will turn on at the previously used speed.
2. On the HMI screen press the box where the speed percentage number is shown. This will bring up a keypad where the desired speed can be entered. The jet has a ramp up function so it may take a couple seconds to reach the desired speed.
3. The speed increase and decrease buttons on the HMI screen adjust the speed in 1% increments each time they are pressed.
4. During the session speed may be increased or decreased by either typing in a speed or by pressing the speed increase and decrease buttons on the HMI screen.
5. To stop the jet there are several ways to do so depending on the reason why the jet is to be stopped.
    a. Jet on/off button on the HMI screen: this is the preferred method if the user just wants to stop the jet but continue the session.
    b. Pause Session button on the HMI screen: this is the preferred method if the user wants to stop the treadmill and jet but continue the session.
    c. Stop Session button on the HMI screen: use this button when the session is over. This button will stop the treadmill and jet and clear the session timer, distance tracker, and jet speed.
6. The jet is programmed to automatically shut off after 1 hour of continuous use. If an extended session is desired, the jet can be restarted immediately.

The direction of the jet stream can be adjusted. To adjust the flow direction, grasp the nozzle and push in the desired direction. The nozzle may be difficult to move at first.

NOTICE The jet nozzle does have a manual flow control. Make sure it is fully open by rotating the nozzle located in the center of the jet counterclockwise.

To use the massage hose:

1. Make sure the jet is off before connecting or disconnecting.
2. Grab the end of the massage hose with the clip-on device.
3. Pull back on the clip and hold while pushing onto the jet nozzle.
4. When fully inserted onto the jet nozzle, push outside ring of the clip toward the jet to lock the massage hose in place.
5. Plug the second jet nozzle with the supplied plug. It attaches the same way the massage hose does. An option to the plug is to rotate the unused jet nozzle closed but that will not fully stop the flow of that nozzle.
6. **NOTICE** Point the end of the hose down into the water to avoid water spraying out of the unit.
7. Follow the instruction above to stop, start, and adjust the flow of the jet.

5.3 Side Single Jet (optional)

The 500 can be equipped with an optional side therapy jet that can be used as resistance for upper body exercises, or it can be used with the massage hose for muscle massage.

To operate the side jet:

1. Press the Jet button on the HMI to turn the jet on.
2. Press the Jet button on the HMI to turn the jet off.
3. To use the massage hose, follow the instructions in the previous section.
4. The jet is programmed to automatically shut off after 1 hour of continuous use. If an extended session is desired, the jet can be restarted immediately.

The flow rate and direction can be varied. To adjust the flow rate, rotate the nozzle located in the center of the jet. Turning the nozzle clockwise will decrease the flow rate. Turning the nozzle counterclockwise will increase the flow rate. To adjust the flow direction, grasp the nozzle and push in the desired direction. The nozzle may be difficult to move at first.

5.4 Variable Water Depth (optional)

The 500 can be equipped with an optional variable water depth system. This system allows the user to reduce the water depth in the pool by up to 8”. To vary the water depth, water is pumped to a temporary storage tank.

To vary the water depth during the session, use the level buttons on the HMI screen. To change water depth, press the level up or down arrows and when the desired depth is achieved, press the level stop button. Adjusting the water level varies the amount of body weight supported by the patient’s lower body. The system does have limiting sensors to prevent overflows.

The water in the storage tank does not have its own water treatment system. **The pool should be kept at full depth except when a user needs the lower water depth.** There is always some water left in the storage tank even when the pool is at full depth. **The water should be transferred to the storage tank and back once per day if the variable water depth hasn’t been used during a session that day.** These steps will maintain a safe water sanitation level in the storage tank and proper function of the pool skimmer.

5.5 Auto Water Fill (optional)

The 500 can be equipped with an optional auto water fill system. This system maintains the water depth of the pool. **The e-stop button must be pressed for automatic filling.**

The water level control switch is located on the control panel in the mechanical space/equipment room. The switch has three positions as described below:

- Manual position: opens the valve to fill the pool. The sensors are disabled and the pool can overflow if left unattended.
- Auto position: automatic filling only occurs when the e-stop button is pressed. When the water level falls below the low sensor for more than 10 seconds, the valve will open and add water until the water level rises above the high sensor for 10 seconds.
- Off position: the system is disabled and won’t fill.

5.6 Video

The 500 is equipped with a video system. There are underwater cameras located at the front and side of the pool to view someone on the treadmill. A TV is used to display the cameras.

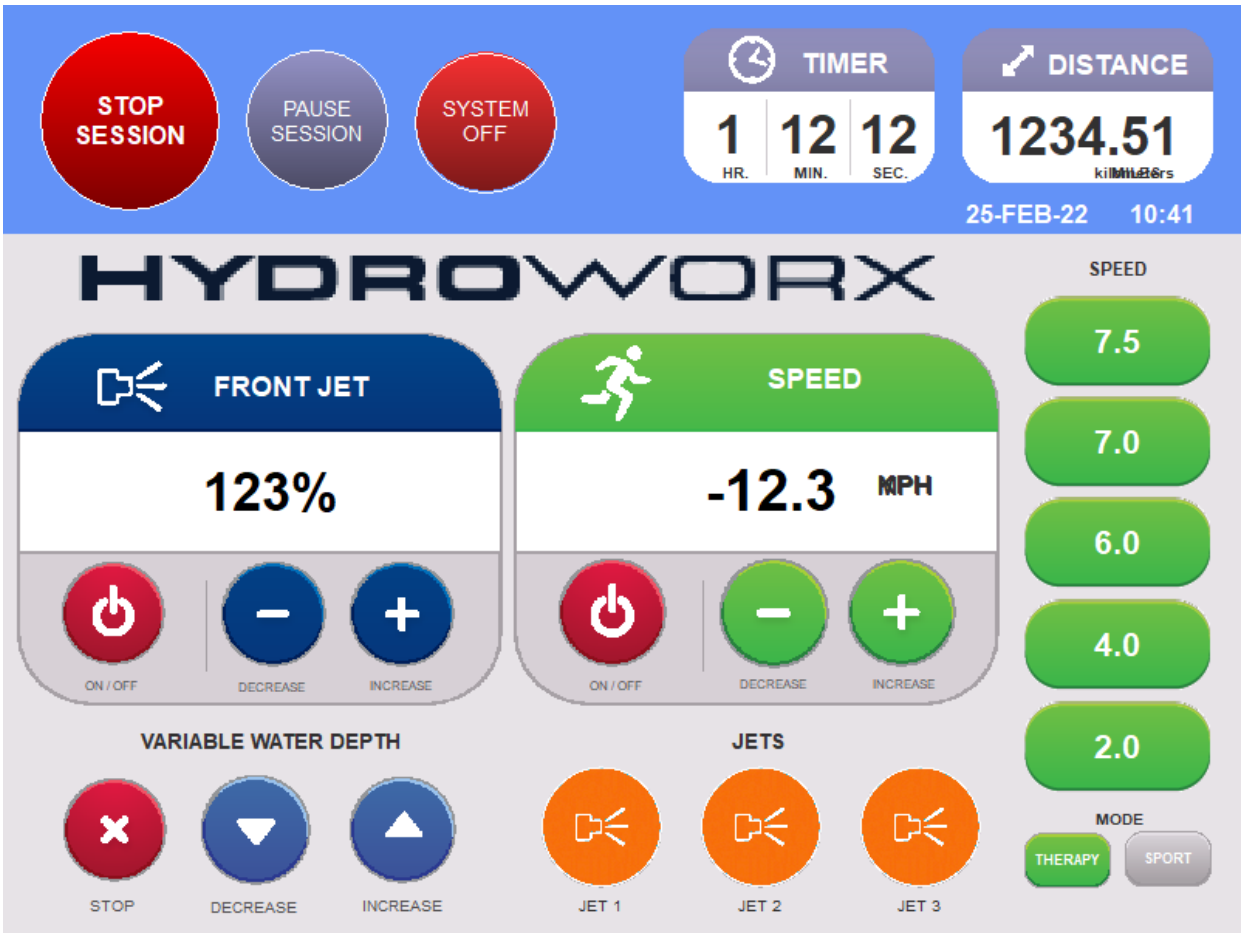
5.7 Heater

The 500 is equipped with an electric heater. The water temperature can be adjusted by pressing the up and down buttons on the heater to the desired temperature. Recommended water temperature is 90°F. It can take a day for the water to heat up when the unit is filled with cold water.

⚠ WARNING Prolonged immersion in hot water may induce hyperthermia. Hyperthermia occurs when the internal temperature of the body reaches a level several degrees above the normal body temperature of 98.6°F. The symptoms of hyperthermia include an increase in the internal temperature of the body, dizziness, lethargy, drowsiness, and fainting. The effects of hyperthermia include failure to perceive heat; failure to recognize the need to exit the unit; unawareness of impending hazard; fetal damage in pregnant women; physical inability to exit the unit; and unconsciousness resulting in the danger of drowning. The use of alcohol, drugs, or medication can greatly increase the risk of fatal hyperthermia.

5.8 HMI

The 500 is equipped with an HMI (Human Machine Interface). The HMI console includes the screen, an emergency-stop button, and a power reset button. The screen is touchscreen and works with wet fingers.



- Stop Session: When pressed this button will stop the jet, stop the treadmill, and clear the timer, distance, and jet speed.
- Pause Session: When pressed this button will stop the treadmill and jet but continue the session.
- System ON/OFF: This is an indicator light showing if the system has control power. GREEN indicates that Control Power is ON. RED indicates that Control Power is OFF. The treadmill and jet are disabled with Control Power OFF.
- Timer: The timer displays the run time of the session. The timer begins when the treadmill or jet is turned on. The timer is paused if the pause session button is pressed. The timer can be manually reset by holding the pause session button for 5 seconds. The timer will automatically reset when a new session begins. (The stop session button must have been pressed at the end of the previous session)
- Distance: This display shows the distance travelled on the treadmill during a session. The distance can be manually reset by holding the pause session button for 5 seconds. The distance will automatically reset when a new session begins.

(The stop session button must have been pressed at the end of the previous session)

- Date and Time: This displays the local date and time.
- Speed: The speed section of the HMI screen shows the speed of the treadmill, turns the treadmill on and off and adjusts speed of the treadmill.
    - The power button turns the treadmill on and off. The treadmill will not start until a speed is selected.
    - The + and – buttons increase or decrease the speed in 0.1mph increments.
    - There are pre-set speeds along the right side for convenience.
    - The Sport/Therapy mode button, if held, will switch between sport and therapy modes. The mode changes the pre-set speeds.
- Jet: The jet section of the HMI screen shows the speed of the jet, turns the jet on and off and adjusts the speed of the jet.
    - The power button turns the jet on and off.
    - If starting a new session, the jet will not start until a speed is selected. If the jet has already been used in the current session the jet will turn on at the previously used speed.
    - Press the box where the speed percentage number is shown. This will bring up a keypad where the desired speed can be entered.
    - The + and – buttons increase or decrease the speed in 1% increments.
- Emergency Stop: Press this button in an emergency to turn control power OFF, disabling the treadmill and jet. To reset, twist the mushroom clockwise ¼ turn and press the power reset button.
- Power Reset: the amber light will be on when the control power is ON. Press the power reset button if the light is off.

5.9 Tablet/TV

The 500 is equipped with a tablet for remote operation. **The tablet is part of the system and should not be used for anything else.** The tablet is a duplicate screen of the HMI. Both the tablet and HMI can be used to control the 500. The tablet does not have any physical buttons such as the emergency stop. A TV is also provided to duplicate the HMI on a larger screen for better visibility to the user. The TV is a display only, it can't perform control functions.

6.0 Maintenance Instructions

6.1 Maintenance quick reference schedule

Daily:

- ✓ Check bromine and pH levels.
- ✓ Test the GFCI device.

Weekly:

- ✓ Check bromine tablet level in the erosion feeder.
- ✓ Shock water with lithium hypochlorite shock.
- ✓ Check alkalinity and calcium hardness levels.
- ✓ Check the filter pressure.
- ✓ Refill unit if not equipped with auto water fill.
- ✓ Wipe down stainless steel.
- ✓ Check treadmill tracking and tension.

Monthly:

- ✓ Drain and thoroughly clean the unit.

6.2 Water Maintenance

Correct water quality is critical for patient health and comfort as well as longevity of the unit. The 500 is supplied with a Taylor water chemistry test kit. This test kit will be used to determine chemical levels and recommendations on how to correct those levels if needed.

6.2.1 Chemicals needed to maintain the water:

- Alkalinity Increaser
- Alkalinity Decreaser-Dry Acid most recommended or Muriatic Acid
- Calcium Hardness Increaser
- Lithium Hypochlorite based Shock
- pH Increaser
- pH Decreaser
- 1" Bromine Tablets
- Chem-Out Chlorine reducer/neutralizer

6.2.2 Potential Issues with Imbalanced Water

- Low Bromine: Inadequate disinfection. Does not kill contaminants properly.

- High Bromine: Corrodes surfaces and equipment. Irritates/ burns user's skin, eyes and respiratory system.
- Low pH: Corrodes surfaces & equipment. Excess sanitizer use. Irritates bather's skin & eyes. Plugs filter. Decrease your sanitizer effectiveness.
- High pH: Scale deposits on surfaces & equipment. May contribute to cloudy water. Poor sanitizer efficiency. Eye and skin irritation. Stains surfaces.
- Low Alkalinity: pH difficult to maintain. Corrosion tendencies. Stains surfaces. Dissolves metals.
- High Alkalinity: pH difficult to adjust, it remains fixed. Cloudy water. Potential for scaling. High pH and low sanitizer efficiency, sanitizer is pH dependent. Potential to plug filter.
- Low Calcium Hardness: May lead to equipment corrosion.
- High Calcium Hardness: May contribute to cloudy water. Scaling of surface, piping, and equipment. Can cause heater inefficiency. Plugs filters. Forms a white film.

6.2.3 Alkalinity

Alkalinity is your pH buffer. It protects your pH from rapid fluctuations. If any of your water chemistry needs adjusting always adjust your alkalinity first. Total alkalinity is the measure of the ability of water to resist changes in pH. Your alkalinity helps level your pH. Often leveling your Alkalinity will balance your pH. Note: You cannot add more than one pound of alkalinity decreaser a day.

Ideal Range (80 to 120 ppm)
Safe Ranges from National Swimming Pool Foundations (60-180 ppm)
Weekly Test Required

6.2.4 Calcium Hardness

Water that contains little to no magnesium is called soft. If you have soft water when you add chemicals into the water, there is nothing for them to hold onto and they won't be effective. There are no chemicals that you can add to decrease your hard water. If your water is too hard you will need to use a water softener when filling the unit.

Recommend Range (250 ppm–350 ppm)
Weekly Test Required

6.2.5 pH

pH is very volatile. Minor issues can greatly change your pH. If your pH is low, then it is acidic. If it is high, then it is basic. Alkalinity and pH work hand in hand with each other.

If one is out of range often the other will follow. Having your pH and Alkalinity in safe ranges is going to increase how effective your sanitizer will be.

Ideal Range (7.2 to 7.6)
Safe Ranges from National Swimming Pool Foundations (pH-7.2-7.8)
Daily Test Required

6.2.6 Bromine

Bromine is the primary sanitizer. To control the Bromine level, adjust the feeder dial on the erosion feeder and the amount of bromine tablets in the erosion feeder. Slightly increasing your dial will increase the bromine but this does take up to 24 hours before the increase is noticed. Increasing the Bromine level is much easier then decreasing the bromine level. The recommended initial setting is to set the dial on the erosion feeder to 1 and fill the feeder 1/3 full. This should last 1 week under low usage. The higher the patient load, the more bromine will be needed so adjust the feeder dial and tablets as needed.

Adding tablets to the erosion feeder process:

1. The feeder can be filled with the recirculation pump on.
2. Turn the dial on the erosion feeder to 0. Make note of where the dial was set so you can return it to the same setting after your done adding tablets.
3. Unscrew the lid on the erosion feeder. 

Wear eye protection and gloves. Do not inhale fumes.
4. Add desired number of tablets. Some water may spill out or use a cup to remove some of the water.
5. Screw the lid back on the erosion feeder.
6. Turn the dial back to the desired setting.

Ideal Range (3.0 to 5.0 ppm)
Safe Ranges from National Swimming Pool Foundations (2-10 ppm)
Daily Test Required

6.2.7 Filter

The 500 is equipped with a cartridge filter and spare cartridge. The filter cartridges are cleanable and reusable. To monitor when the filter needs cleaned, look at the pressure gauge on top of the filter. When the pressure increases 10 psi above a clean filter, the filter needs cleaned. If the filter pressure increases too much it will restrict the flow of the recirculation system and it won't function properly. It may also cause damage to the recirculation components.

Filter cleaning process:

1. Turn off power to the control panel.
2. Close the isolation valves on each side of the filter.
3. Open the valve to the drain port on the filter and open the bleeder screw on top of the filter.
4. Unscrew the lid on the filter and remove the filter cartridge.
5. Install a clean filter cartridge and screw the lid back on the filter.
6. Close the drain valve and open the isolation valves.
7. Air will begin to come out of the bleeder on the filter. When a steady flow of water is coming out of the bleeder, the bleeder can be screwed closed.
8. Turn the power back on to the control panel.
9. Bleed the air out of the erosion feeder by unscrewing the lid until water reaches the top of the feeder.
10. The unit is now operational again.
11. To clean the used filter cartridge, soak in a tub with a pool filter cleaning solution such as Filter Cleanse. Follow cleaner manufacturer's directions.

6.3 Cleaning

The 500 should be wiped down weekly and thoroughly cleaned monthly. The 500 is made of acrylic with stainless steel accessories. These materials are corrosion resistant but proper cleaning is needed to maintain a like new appearance.

6.3.1 What to use:

- Mild detergent such as dish soap
- Plastic cleaner such as Novus #1 or Brillianize
- Clean, soft cotton or microfiber cloth
- 10-20% vinegar solution for scale buildup when needed
- Stainless steel cleaner such as Spectra-Clean system 1, 3M, and Weiman

6.3.2 What not to use:

- Scour pads, scotch-brite pads, steel wool, sandpaper
- Paper towels
- Ammonia based cleaners like multipurpose kitchen/bathroom cleaners or Windex
- Alcohol based cleaners
- Bleach

6.3.3 Weekly cleaning

Every week the areas above the water line should be wiped down. Use a plastic cleaner or mild detergent to clean plastics and a stainless steel cleaner to clean the metal

handrails, and pool signs. If the stainless steel starts to stain, then it should be cleaned more often.

6.3.4 Monthly Cleaning

Every month the pool should be drained, thoroughly cleaned, and refilled with fresh water. The frequency of this routine cleaning can be adjusted based on patient load.

1. Turn power off to the 500.
2. Drain the pool by opening the drain valve.
3. Use a wet/dry vac and/or a sump pump to remove the rest of the water from the bottom of the pool.
4. Thoroughly clean the entire inside of the pool with a mild detergent and soft cloth. Clean the treadmill belt as well.
5. Rinse out the pool.
6. Close the drain valve.
7. Refill the pool with fresh water up to the center of the skimmer.
8. Bleed air from the filter and erosion feeder.
9. Turn power back on to the 500.
10. Follow water maintenance instructions to treat the water.

6.4 Treadmill Belt

The treadmill belt requires regular tracking and tensioning as the belt ages. There are tensioning screws located at the rear corners of the treadmill. Turning the screws counterclockwise will tighten the belt. HydroWorx supplies a T-handle tool to adjust the screws. To track the belt, if the belt has moved to the left than turn the screw on the left part of a turn. Run the treadmill at 3mph for about 30 seconds and see if the belt has moved back to center. Make adjustments as needed to move the belt to center. If the belt moves too far than undo the last adjustment part way. The belt may stray slightly from center throughout the speed range and that is normal.

To check the treadmill belt tension, stand on the treadmill and turn it on at 2mph. If you notice any slippage then increase the belt tension. Turn each screw 1 turn and check the tension again until no slippage occurs. **NOTICE** Do not over tension the belt as that may cause damage or a motor overload fault.

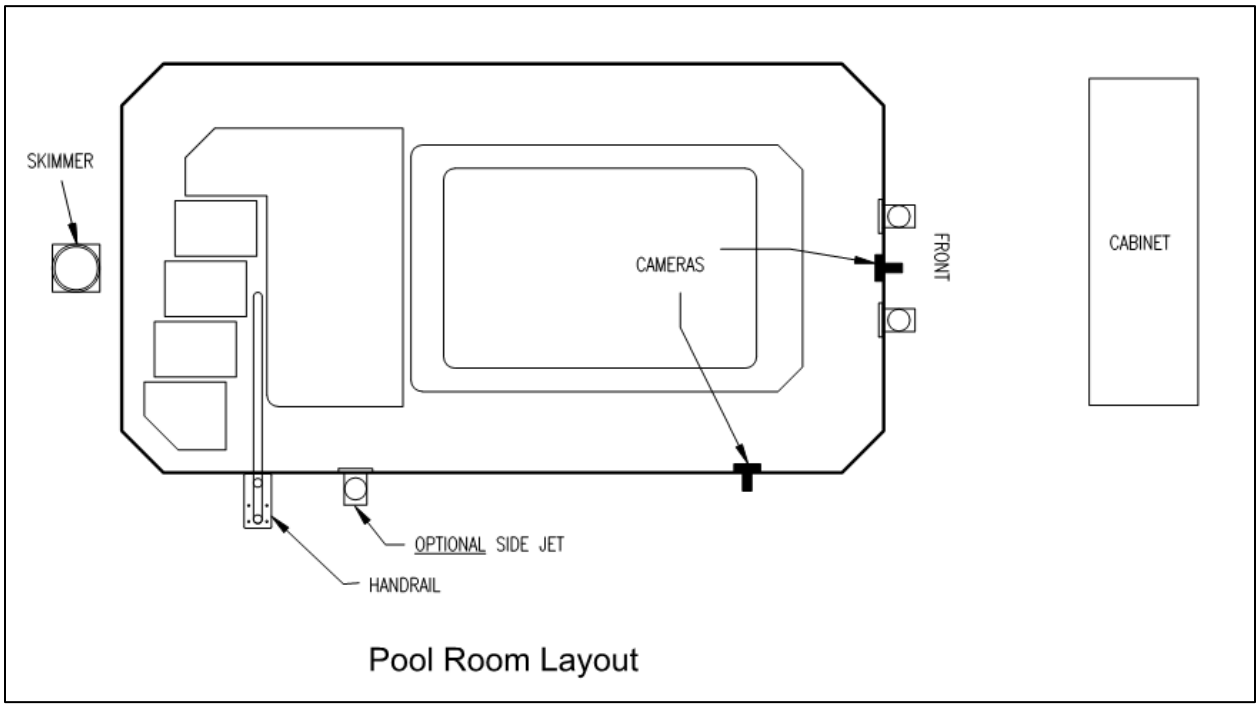
6.5 GFCI

The 500 is equipped with GFCI protection. This device shall be tested before every use to verify they are operating correctly. **⚠ DANGER** **Risk of electric shock or death. If**

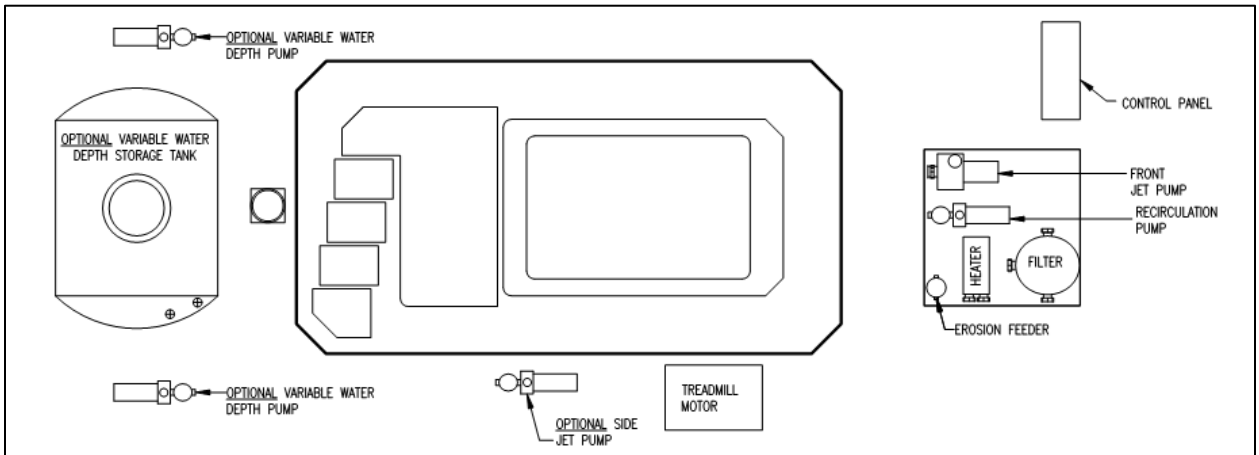
the device is not operating correctly turn off power to the unit and do not operate until it is fixed by HydroWorx. There is a GFCI test switch on the front top left corner of the control panel.

To test the GFCI device, turn the small black test/reset switch to test for a few seconds. The main breaker should trip, and the unit will lose power. If the breaker trips then the device is working properly, and you can reset the unit. To reset the unit, turn the small black test/reset switch to reset for a few seconds. Turn the large black main breaker switch in the upper right corner to reset and then to on. Press the amber power reset button to restore control power to the unit.

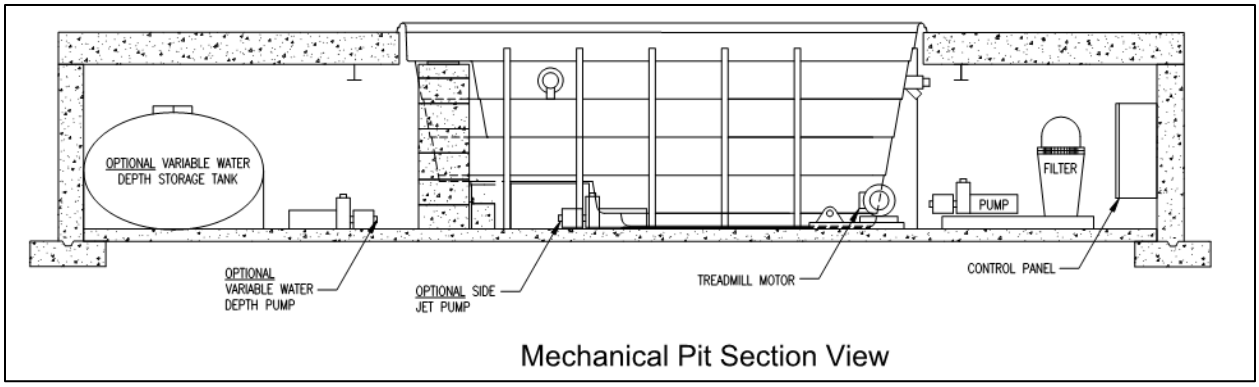
Appendix A – Drawings and Illustrations



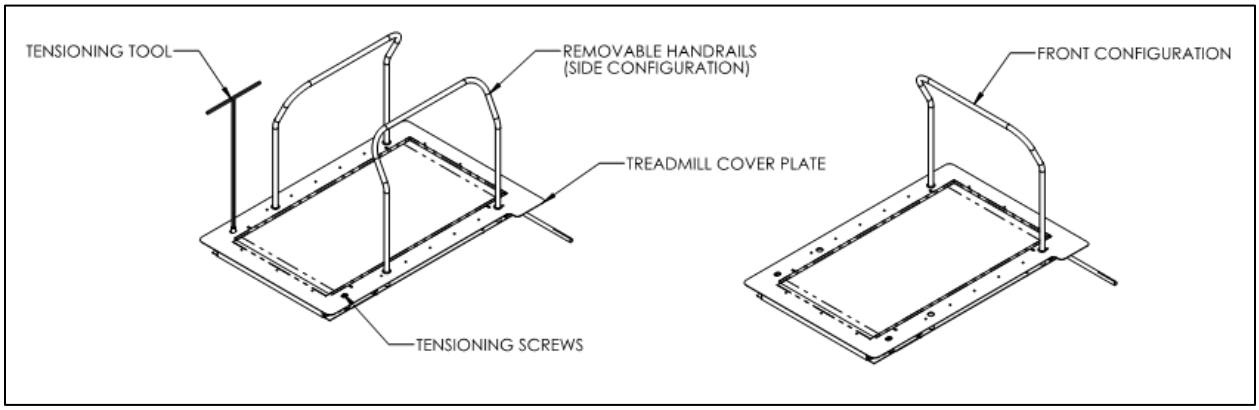
Pool Room Layout

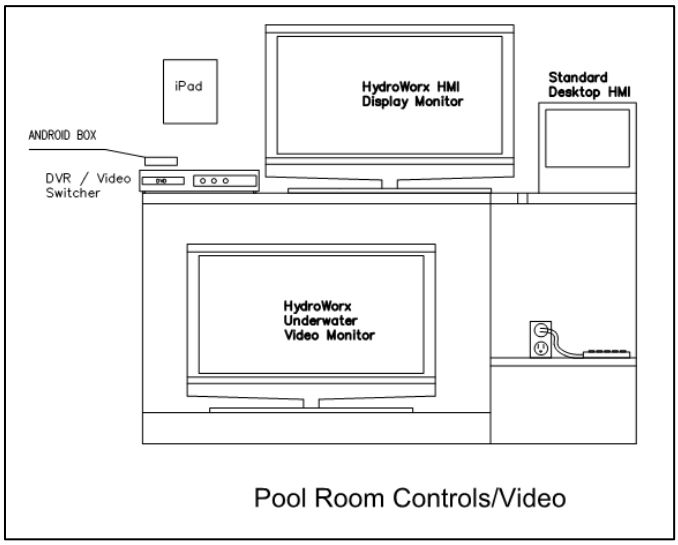


Mechanical Pit Layout



Mechanical Pit Section View





Pool Room Controls/Video